M. Riley Owens

ABOUT ME

I am a first-year graduate student in the Department of Astronomy at the University of California, Berkeley, where I work with Prof. Dan Stark. My research interests focus on understanding the evolution and physics of galaxies in the early universe, from the end of reionization to the first galaxies.

EDUCATION

Doctor of Philosophy in Astrophysics 2025-
University of California, Berkeley

Advised by Prof. Daniel Stark | GPA: 3.87

Bachelor of Science in Astrophysics and Physics, Minor in Mathematics 2018-2022
University of Cincinnati

Advised by Prof. Matthew Bayliss | GPA: 3.73

POSITIONS

Graduate student researcher 2025-
University of California, Berkeley

Postbaccalaureate researcher 2022
West Virginia University

Postbaccalaureate researcher 2022-2025
University of Cincinnati

NSF REU student 2021
West Virginia University

Undergraduate researcher 2019-2022
University of Cincinnati

AWARDS AND HONORS

NSF GRFP Honorable Mention, National Science Foundation 2025
Fulbright Semi-Finalist, Fulbright Program 2023
Chambliss Astronomy Achievement Award, American Astronomical Society 2023
Dean's List, University of Cincinnati 2018-2022
University Honors Scholar, University of Cincinnati 2018-2022

GRANTS, SCHOLARSHIPS, AND FELLOWSHIPS

Undergraduate Research Fellowship, University of Cincinnati $750 2022
Cincinnatus Scholarship, University of Cincinnati $X? 2018-2022

PUBLICATIONS

For a complete record of my publications and associated statistics, please visit my ORCID record on NASA ADS.

FIRST AUTHOR

Connecting Lya and Ionizing Photon Escape in the Sunburst Arc, 5 citations 2024
The Astrophysical Journal

DOI: 10.3847/1538-4357/ad9247

M. R. Owens, K. J. Kim, M. B. Bayliss, T. E. Rivera-Thorsen, K. Sharon, J. R. Rigby, A. Navarre, M. Florian, M. D. Gladders, J. G. Burns, G. Khullar, J. Chisholm, G. Mahler, H. Dahle, C. M. Malhas, B. Welch, T. A. Hutchison, R. Gassis, S. Choe, P. Adhikari

COAUTHOR

Lyman Continuum escaping from in-situ formed stars in a tidal bridge at z=3, arXiv — 2025

DOI: 10.48550/arXiv.2512.19967

T. E. Rivera-Thorsen, A. Le Reste, M. J. Hayes, S. Flury, A. Saldana-Lopez, B. Welch, S. Choe, K. Sharon, K. Kim, **M. R. Owens**, E. Solhaug, H. Dahle

JWST & the Waz Arc I: Spatially Resolving the Physical Conditions within a Post-Starburst Galaxy at Redshift 5 with NIRSpec IFS, arXiv — 2025

DOI: 10.48550/arXiv.2512.02000

T. A. Hutchison, G. Khullar, J. R. Rigby, B. Welch, M. K. Florian, K. Sharon, I. Sierra, J. Sarmiento, G. Mahler, N. J. Cleri, R. Bezanson, M. D. Gladders, M. B. Bayliss, J. S. M. Karp, D. Berry, A. Ross, T. E. Rivera-Thorsen, S. C. Choe, H. Dahle, J. Chisholm, E. L. Lambrides, R. L. Larson, G. M. Olivier, **M. R. Owens**, E. Solhaug

Tracing Structure: Shape and Centroid Deviations in 39 Strong Lensing Clusters as a Test of Cluster Formation Predictions, arXiv — 2025

DOI: 10.48550/arXiv.2511.22661

R. Gassis, M. B. Bayliss, K. Sharon, G. Mahler, M. D. Gladders, M. McDonald, H. Dahle, M. K. Florian, J. R. Rigby, L. A. Elicker, **M. R. Owens**, P. Adhikari, G. Khullar

The Chandra Strong Lens Sample: Measuring the Dynamical States and Relaxation Fraction of a Sample of 28 Strong Lensing Selected Galaxy Clusters, arXiv — 1 citations — 2025

DOI: 10.48550/arXiv.2511.12707

R. Gassis, M. B. Bayliss, M. McDonald, K. Sharon, G. Mahler, M. D. Gladders, H. Dahle, M. K. Florian, J. R. Rigby, L. A. Elicker, **M. R. Owens**, P. Adhikari, G. Khullar

The Sunburst Arc with JWST. IV. The importance of interaction, turbulence, and feedback for Lyman continuum escape, arXiv — 2 citations — 2025

DOI: 10.48550/arXiv.2510.11702

T. E. Rivera-Thorsen, B. Welch, T. Hutchison, M. J. Hayes, J. R. Rigby, K. Kim, S. Choe, M. Florian, M. B. Bayliss, G. Khullar, K. Sharon, H. Dahle, J. Chisholm, E. Solhaug, **M. R. Owens**, M. D. Gladders

Cool Dark Gas in Cygnus X: The First Large-scale Mapping of Low-frequency Carbon Recombination Lines, The Astrophysical Journal — 1 citations — 2025

DOI: 10.3847/1538-4357/adfa17

K. L. Emig, P. Salas, L. D. Anderson, D. A. Roshi, L. Bonne, A. D. Bolatto, I. A. Grenier, R. C. Levy, D. J. Linville, M. Luisi, **M. R. Owens**, J. Poojapriyatharsheni, N. Schneider, L. Tibaldo, A. G. G. M. Tielens, S. K. Walch, G. J. White

COOL-LAMPS. VIII. Known Wide-separation Lensed Quasars and Their Host Galaxies Reveal a Lack of Evolution in M$_{BH}$/M$_{\star}$ Since $z \sim 3$, The Astrophysical Journal — 6 citations — 2025

DOI: 10.3847/1538-4357/addabf

A. P. Cloonan, G. Khullar, K. A. Napier, M. D. Gladders, H. Dahle, R. Rosener, J. Sullivan, M. B. Bayliss, N. Chicoine, I. Escapa, D. Garza, J. Garza, R. Glusman, K. Gozman, G. Horwath,

A. Kisare, B. C. Levine, O. Liang, N. Malagon, M. N. Martinez, A. Masegian, O. S. Matthews Acuña, S. D. Mork, K. Niu, **M. R. Owens**, Y. Pan, J. R. Rigby, K. Sharon, I. Sierra, A. A. Stark, E. Sukay, M. Tagliavia, M. Tamargo-Arizmendi, K. Tavangar, R. Teixeira, K. Tsiane, R. Tu, G. Wagner, E. A. Zaborowski, Y. Zhang, Y. "M." Zhao

The Sunburst Arc with JWST: II. Observations of an Eta Carinae analog at z=2.37, Astronomy & Astrophysics 18 citations 2025

DOI: 10.1051/0004-6361/202450685

S. Choe, T. E. Rivera-Thorsen, H. Dahle, K. Sharon, **M. R. Owens**, J. R. Rigby, M. B. Bayliss, M. J. Hayes, T. Hutchison, B. Welch, J. Chisholm, M. D. Gladders, G. Khullar, K. Kim

The Sunburst Arc with JWST. III. An Abundance of Direct Chemical Abundances, The Astrophysical Journal 30 citations 2025

DOI: 10.3847/1538-4357/ada76c

B. Welch, T. E. Rivera-Thorsen, J. R. Rigby, T. A. Hutchison, G. M. Olivier, D. A. Berg, K. Sharon, H. Dahle, **M. R. Owens**, M. B. Bayliss, G. Khullar, J. Chisholm, M. Hayes, K. J. Kim

The Sunburst Arc with JWST: I. Detection of Wolf-Rayet stars injecting nitrogen into a low-metallicity, z=2.37 proto-globular cluster leaking ionizing photons, Astronomy & Astrophysics 42 citations 2024

DOI: 10.1051/0004-6361/202450359

T. E. Rivera-Thorsen, J. Chisholm, B. Welch, J. R. Rigby, T. Hutchison, M. Florian, K. Sharon, S. Choe, H. Dahle, M. B. Bayliss, G. Khullar, M. Gladders, M. Hayes, A. Adamo, **M. R. Owens**, K. Kim

COOL-LAMPS. VI. Lens Model and New Constraints on the Properties of COOL J1241+2219, a Bright z=5 Lyman Break Galaxy and its z=1 Cluster Lens, The Astrophysical Journal 4 citations 2024

DOI: 10.3847/1538-4357/ad22de

M. Klein, K. Sharon, K. Napier, M. D. Gladders, G. Khullar, M. Bayliss, H. Dahle, **M. R. Owens**, A. Stark, S. Brownsberger, K. J. Kim, N. Kuchta, G. Mahler, G. Smith, R. Walker, K. Gozman, M. N. Martinez, O. S. Matthews Acuña, K. Merz, J. A. Sanchez, D. J. K. Stein, E. O. Sukay, K. Tavangar

Resolving Clumpy versus Extended Lya in Strongly Lensed, High-redshift Lya Emitters, The Astrophysical Journal 4 citations 2024

DOI: 10.3847/1538-4357/ad10ad

A. Navarre, G. Khullar, M. B. Bayliss, H. Dahle, M. Florian, M. Gladders, K. J. Kim, **M. R. Owens**, J. Rigby, J. Roberson, K. Sharon, T. Shibuya, R. Walker

Small Region, Big Impact: Highly Anisotropic Lyman-continuum Escape from a Compact Starburst Region with Extreme Physical Properties, The Astrophysical Journal Letters 44 citations 2023

DOI: 10.3847/2041-8213/acf0c5

K. J. Kim, M. B. Bayliss, J. R. Rigby, M. D. Gladders, J. Chisholm, K. Sharon, H. Dahle, T. E. Rivera-Thorsen, M. K. Florian, G. Khullar, G. Mahler, R. Mainali, K. A. Napier, A. Navarre, **M. R. Owens**, J. Roberson

Strongly lensed [O III] emitters at Cosmic Noon with Roman: Characterizing extreme emission line galaxies on star cluster complex scales (100 pc), arXiv 2023

DOI: 10.48550/arXiv.2307.01247

K. J. Kim, M. B. Bayliss, H. Dahle, T. Hutchison, K. Sharon, G. Mahler, **M. R. Owens**, J. E. Rhoads

The Connection between Galactic Outflows and the Escape of Ionizing Photons, The Astrophysical Journal — 41 citations — 2022

DOI: 10.3847/1538-4357/ac9cd6

R. Mainali, J. R. Rigby, J. Chisholm, M. Bayliss, R. Bordoloi, M. D. Gladders, T. E. Rivera-Thorsen, H. Dahle, K. Sharon, M. Florian, D. A. Berg, S. Sharma, **M. R. Owens**, K. Kjellgren, K. J. Kim, J. Wayne

OBSERVATIONS

PRINCIPAL INVESTIGATOR

Resolving the Lyman-alpha kinematics of a lensed galaxy with a rare Lyman-alpha profile, Gemini/GMOS-N — 7 hr — Gemini 2024B FT

R. Owens, M. Bayliss, E. Solhaug, G. Khullar, B. Welch, K. Kim, M. Gladders, E. Rivera-Thorsen, H. Dahle

Identifying galaxy-lensed Lyman-alpha emitters, Gemini/GMOS-S — 2 hr — Gemini 2023A FT

R. Owens, K. Kim, M. Bayliss, H. Dahle, J. Burns, K. Sharon, G. Smith, M. Klein, N. Kuchta, R. Walker, E. Rivera-Thorsen, G. Mahler, G. Khullar

COINVESTIGATOR

Spatially Resolving Highly Ionized Channels of Lyman Continuum and Lyman Alpha Escape on 10's of Parsecs Scales In A Strongly Lensed Galaxy, HST/ACS, HST/WFC3 — 44 orbits — HST Cycle 32 GO

M. Bayliss, H. Dahle, M. Florian, M. D. Gladders, T. A. Hutchison, G. Khullar, K. J. Kim, **M. R. Owens**, J. R. Rigby, T. E. Rivera-Thorsen, K. Sharon

Time delay cosmography with strong cluster lenses, HST/ACS, HST/WFC3, JWST/NIRCam — 14 orbits, 12 hr — HST Cycle 32 GO

H. Dahle, M. Bayliss, A. P. Cloonan, M. Florian, M. D. Gladders, T. A. Hutchison, G. Khullar, K. J. Kim, G. Mahler, K. Napier, **M. R. Owens**, J. R. Rigby, T. E. Rivera-Thorsen, K. Sharon, B. Welch

Spatially Resolved Physical Conditions In a Strongly Lensed X-ray Emitting Dwarf Starburst at Cosmic Noon, Magellan/FIRE — 21 hr — NOIRLab 2024B

M. Bayliss, J. Roberson, H. Dahle, M. Gladders, G. Khullar, K. Kim, **R. Owens**, K. Sharon

Resolving Multi-Peaked Lyman-alpha In A Strongly Lensed Galaxy at z=4: Revealing A Candidate Highly Magnified LyC Leaking Galaxy, Gemini/GMOS-N — 3 hr — Gemini 2024A FT

M. Bayliss, A. Navarre, K. Sharon, H. Dahle, **R. Owens**, M. Gladders, K. Kim

A Dynamical Study of an HST+Chandra Sample of Strong Lensing Clusters, Gemini/GMOS-N, Gemini/GMOS-S — 28 hr — Gemini 2024A

M. Bayliss, R. Gassis, P. Adhikari, M. Gladders, H. Dahle, K. Napier, G. Khullar, G. Mahler, **R. Owens**

The Partially Ionized Medium around HII Regions - the low frequency complement, GBT/PF1 342 MHz — 49 hr — GBT 23B Regular

P. Salas, K. Emig, **M. Owens**, D. A. Roshi, L. Anderson

Galaxies Under Construction: Resolved Scaling Relations and Stellar Mass Assembly as Revealed by Lensed Star-Forming Clumps at Cosmic Noon, JWST/NIRCam, JWST/NIRSpec | 68 hr | JWST Cycle 2 GO

M. Florian, R. Bezanson, J. Chisholm, H. Dahle, M. Gladders, T. Hutchison, G. Khullar, K. Kim, G. Mahler, A. Navarre, **R. Owens**, J. Rigby, T. Rivera-Thorsen, J. Roberson, K. Sharon, I. Shivaei, B. Welch, K. Whitaker